

Learning target AD5, version 1

Draw annotated number lines and apply Derivative Tests to classify the critical points as local extrema.

Consider the function $f(x) = 2x^3 + 24x^2 - 54x + 7$.

1. Find the critical points by hand using algebra, and draw a sign chart (aka labeled number line).
2. Restricting the domain of this function to the interval $[-8, 2]$, what are the absolute maximum and absolute minimum values of this function, and where do they occur?
3. What if we change the domain to $[-10, 2]$?